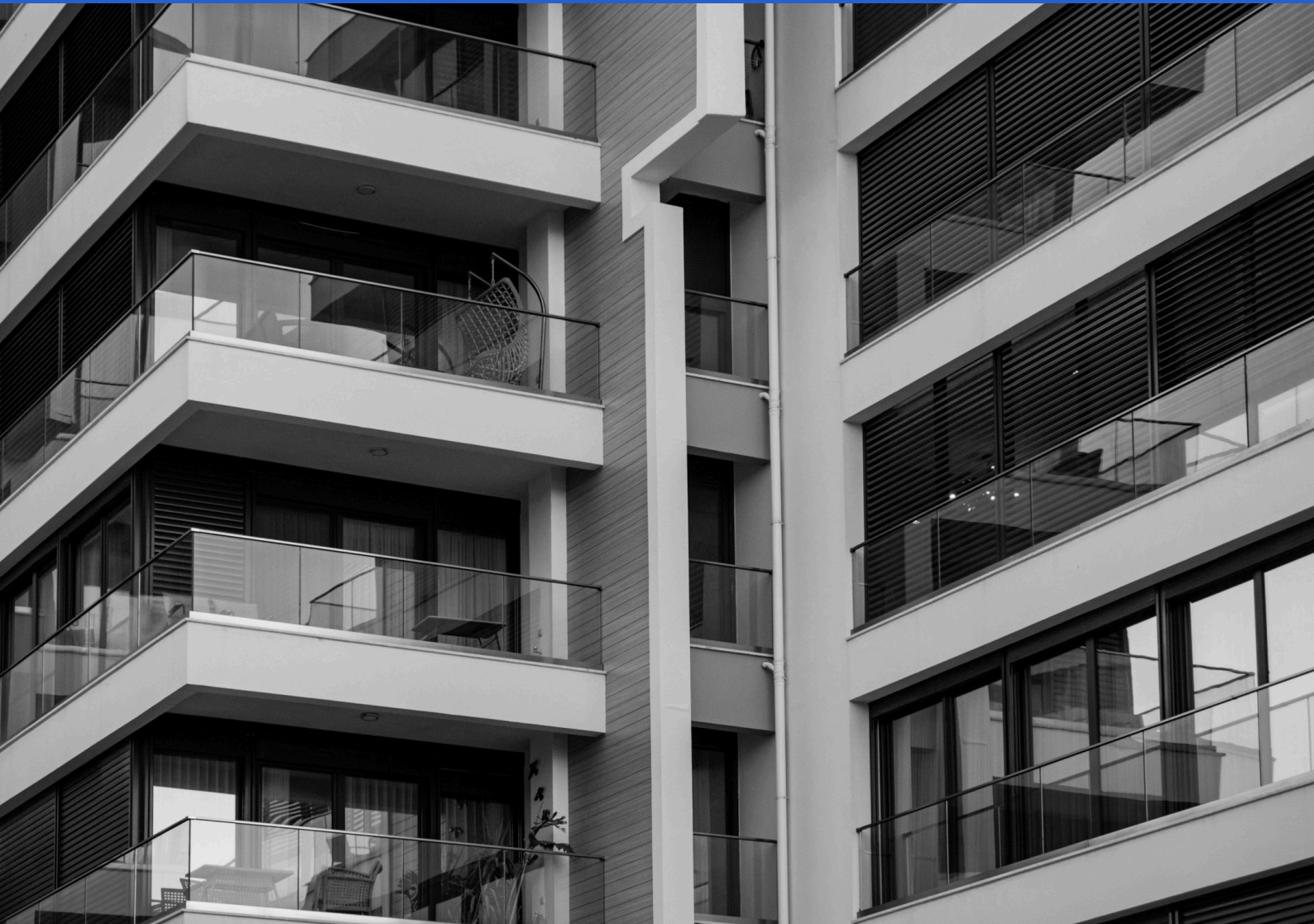


PRESENTED BY:
GROUP 8

SUPERVISOR:
MR. THABISO LUNGA APHANE



PLANNING PHASE

VALORA

PROPERTY INTELLIGENCE AND PRIVATE
COMMUNITY MANAGEMENT HUB

Table of Contents

Identification of Need: 3

Preliminary Investigation: 3

Feasibility Study: 4

Project Planning: 5

Project Scheduling (PERT Chart and Gantt Chart): 6

Software Requirements Specification: 8

Data Models: 10

Identification of Need:

It can be difficult for the people of South Africa to find worthwhile information about a property or to have a sense of control within their community, regardless of whether they are looking to purchase or already reside within it. The pre-existing property sites like Property24 are primarily driven towards listing properties and do not provide much information about the potential future success or profitability of a property, such as what the property could earn and the property value. There is also a need for the people in these communities to be able to report problems, monitor repairs, locate a reliable service provider or converse with the HOA. Sometimes communities lack a physical location to have meetings, which causes delays in issue resolution. This leads to buyers and investors making decisions without being fully aware of all the issues within the community. The VALORA system is attempting to solve this issue by arranging the information on properties, tracking complaints, providing lists of service providers, as well as tools for homeowner associations in one location. This will help individuals to make decisions more easily and for communities to function effectively.

Preliminary Investigation:

The preliminary investigation took place to establish if there is a South African property management problem that is being solved by the proposed VALORA system, and if there's enough motivation for the system's continuation. The investigation's attention was directed toward examining the challenges property owners, investors, HOAs and residents of gated communities encounter, as well as the availability and market demand for this technology. The investigation involved a literature review of the Property24 platform, the Private Property platform, the levies platform, and the South African property technology landscape. It also involved informal interviews with residents and HOA administrators to validate user pain points and technology assessment, in order to determine the technical readiness of AI/ML tools to support property values.

A key part of the investigation was to understand if there was a market opportunity for a platform that delivers property valuation intelligence with community management capabilities. The key findings were that none of the platforms currently offer AI valuation along with complaint tracking, residents need faster and more streamlined solutions to complaints, international platforms like Zillow offer valuations but lack South African data and HOA management features, and property sales data can be obtained from deeds offices and property practitioners. The investigation has confirmed the original problem statement being that there is a lack of investment intelligence and effective community management systems in the current property management and listing systems in SA, for private housing complexes and gated communities. There is also no central announcements, maintenance updates and community solutions platform. Most complaints are made via WhatsApp, with no accountability and issues never being resolved. The difficulties diminish the effectiveness of community management and property investment decision-making.

From the results of the preliminary study, it is recommended to move this project to the feasibility study phase. The priority should be creating an MVP that incorporates AI-based property valuation features. Thereafter, the product can then be piloted into a specific community and subsequently tested with partner property managers of HOA-managed estates. Compliance with the POPI Act should also be ensured from the start of development. Overall, the study establishes that VALORA is a valid solution that is needed in the South African property market.

Feasibility Study:

Technical Feasibility:

The VALORA project is technically feasible because the technologies needed to develop the system are easily available and well supported. The team intends to use Python/FASTAPI for the backend, React and JavaScript for the frontend, PostgreSQL for data storage and AWS for hosting the application. These technologies are widely utilized in today's modern web development. Existing ML tools such as Scikit-learn and XGBoost will also be used to develop the AI features for property valuation and the categorization of complaints. Due to our team members being assigned different roles (frontend and backend development, databases, and documentation), the project therefore has the necessary skills and resources to be successfully implemented.

Operational Feasibility:

It is feasible to operate, as it addresses the practical issues and concerns of residents, property buyers, investors, estate agents and HOAs. The goal of this system is to provide property intelligence features and tools to report complaints, track issues, and identify trusted service providers. This platform will be designed to suit various user groups, with their specific functions and permissions, allowing for practical, easy use. Using Agile Scrum enables frequent feedback from the user, which will help the team refine the system during development.

Economic Feasibility:

VALORA is economically feasible because many of the technologies being used are open source and do not require expensive software licenses, which help keep development costs low. AWS supports the team with a cost-effective system, as they just need to pay for what is used. This platform offers value to the property buyers, investors, residents and estate agents as it provides both information and services in a unified platform. It also helps in making better property decisions, improved community management and future revenue opportunity, however, developing and maintaining the system would incur costs.

Project Planning:

Project Scope:

Within the scope there will be three main parts: the property module, community module, and user management. The property module will include property listings, the ability to search, AI valuation, rental yield calculation and ROI forecasting. Next is the community module which will include the tracking of complaints, service provider directories, HOA announcements and the resident's notifications. Finally, user management which will have the authentication, authorization and profile management. Outside of the scope will contain transactions regarding the purchasing of properties, banking integration, mortgage applications and legal conveyancing services.

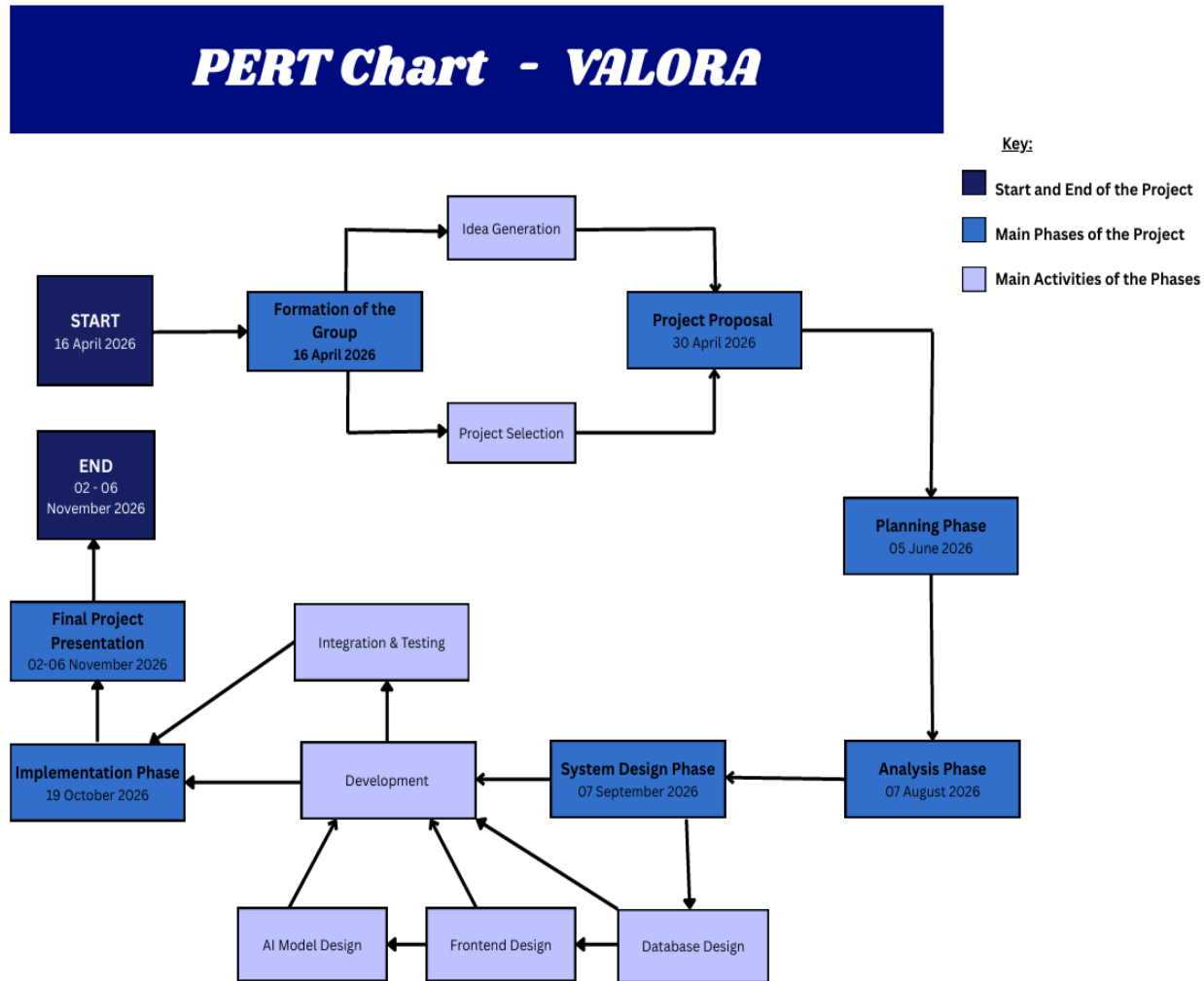
Software Resources:

The software resources would include Python, FastAPI, React, PostgreSQL, AWS, GitHub, Jira and Figma.

Phases of Planning:

- **Requirement Analysis** - This phase involves understanding what the customer needs, what features they would like and finding a start point.
- **System Design** - The second phase involves creating the Gantt and PERT charts, as well as the Data Models, which help to determine what will be designed and when.
- **Development** - The web application will be designed and split into 2 parts: backend and frontend. The backend being made using python with FastAPI and the frontend being made using a combination of JavaScript and React.
- **Testing** - All the functional and non-functional requirements will be tested. This will find any errors or space for improvement.
- **Deployment** - VALORA will then be deployed. It will be accessible through both mobile devices and desktops.
- **Review and Documentation** - The web application will be inspected, and all documentation of VALORA will be established.

Project Scheduling (PERT Chart and Gantt Chart):



Gantt Chart - VALORA

Project Name:	VALORA - Gantt Chart
Project Start Date:	2026/04/16
Project End Date:	2026/11/02
Current Date:	2026/06/03
Weeks in Progress:	WK 7

[illegible]

Software Requirement Specification:

Functional Requirements (FR):

- **Property Intelligence Module:**

The system should be secured with unique credentials and dynamically restrict features available to each user based on five different roles: Casual Visitor, Resident, Investor, Estate Agent and HOA Admin. It should include automated ML valuation, where the system should take in user inputs of property parameters (unit square footage, number of bedrooms/bathrooms, localized suburb name) and return a real time valuation estimate based on the built-in Scikit-learn/XGBoost model. For registered Investors, premium investor analytics provide the mathematical projections that are calculated and outputted to include a 10-year ROI curve and estimated rental yield percentages. The estate agent listing interface should include a Create, Read, Update and Delete (CRUD) interface for verified estate agents to add, view, edit and remove a property listed within the community geographical limits.

- **Community Management Module:**

Resident complaint logging & tracking must be able to be used by a registered resident to log an infrastructure or community issue, upload images as evidence and see the status (Logged, In Progress, Resolved) updated in real time. AI Natural Language Classification needs to analyse the content of the resident's reported complaint and apply a text classification category tag (Plumbing, Security, Electrical, etc.) before the complaint is manually reviewed by an administrator. The system needs to provide a secure dashboard for HOA admins to moderate incoming complaints, update tracking statuses and send community-wide alerts or notices. Verified Service Provider Directory enables the HOA admin to add, subtract and verify entries to the list of local service providers that can be displayed to the residents of the community. The system shall enable residents to give the service providers a rating (1-star to 5-star and a written review) after they have been hired.

Non-Functional Requirements (NFR):

- **Performance:**

With FastAPI Core Latency, the Python backend will respond to complicated analytical queries and deliver machine learning evaluation scores within 2 seconds when traffic peaks. The user interface to be developed in React should be designed to optimize the rendering of the resources so that a complete page can be displayed within 3 seconds on average on a standard mobile and desktop connection in South Africa.

- **Security and Availability:**

The application should implement role-based access tokens with JSON Web Tokens (JWT) to be used on all API requests, and sensitive resident profiles and data fields must be encrypted at rest in the database. The platform architecture should ensure a 99.5% deployment uptime, with the ability to scale up automatically when there is an

unexpected traffic surge, via AWS auto-scaling configurations.

- **Usability:**

The website should be fully responsive (works on desktop, mobile devices). The system should contain consistent navigation and be accessible to users with basic computer literacy. All complaint forms are to have labels and a message when there is an error.

- **User Interfaces:**

Visitor View includes searching and displaying visitor valuations for public property. Resident Dashboard provides access to complaint form, complaint list, service directory, and announcements. Investor View offers comprehensive investment reports and property short listings. Agent Dashboard displays listing management and inquiry log. The Complaint moderation, provider management and analytics features are available in HOA Admin Panel.

- **Data Source:**

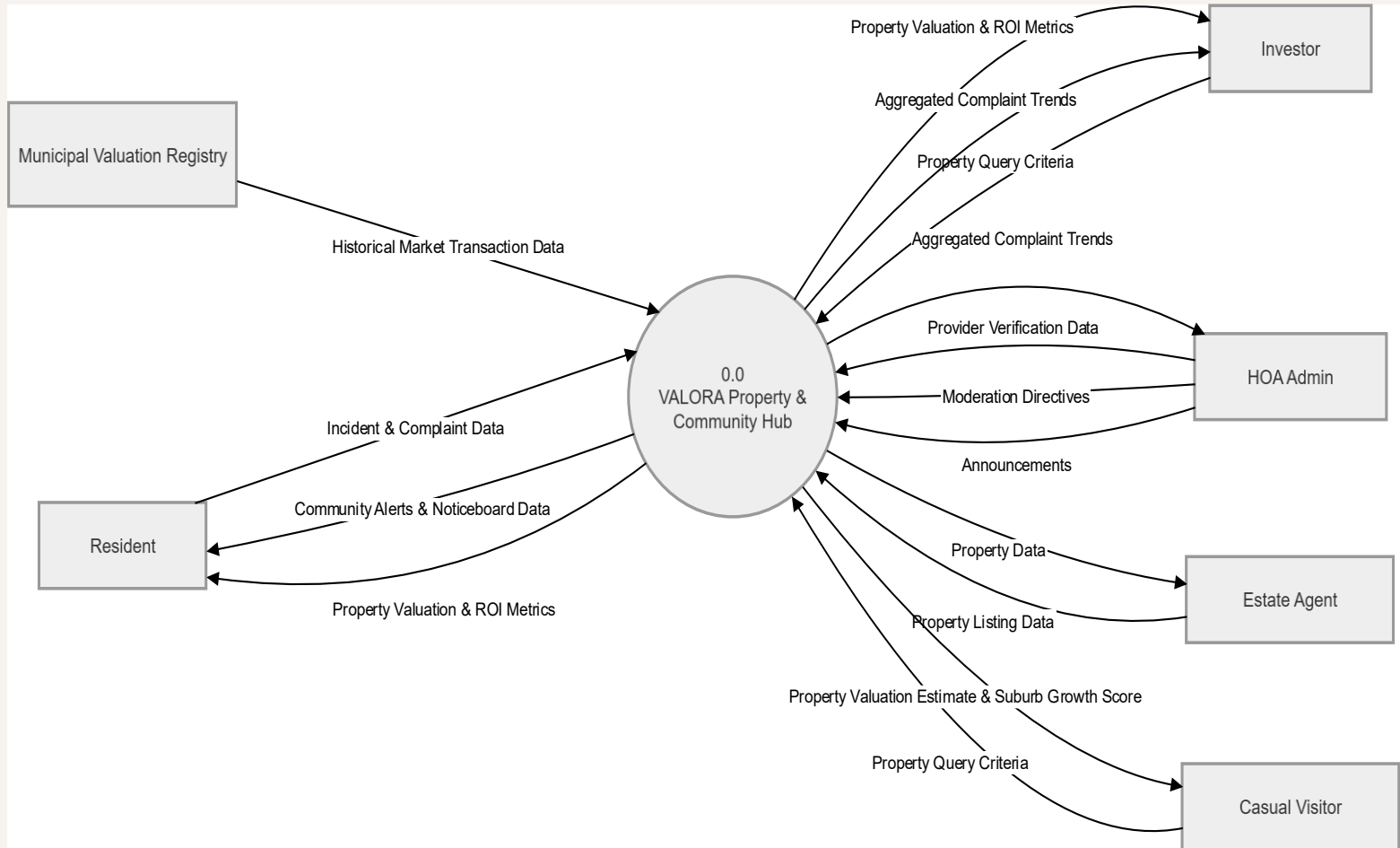
HOA should provide all property data (units, sizes, sales history) in a spreadsheet or API. There does not need to be any external property portals scraped.

- **Constraints:**

The platform is to be delivered by November 2026. It is available for use at a single housing complex only (not available for use at multiple complexes in MVP). The development team is made up of nine specializing members. There is no external money, and hosting costs must either be low cost or free (Hetzner low-cost VPS, AWS free tier).

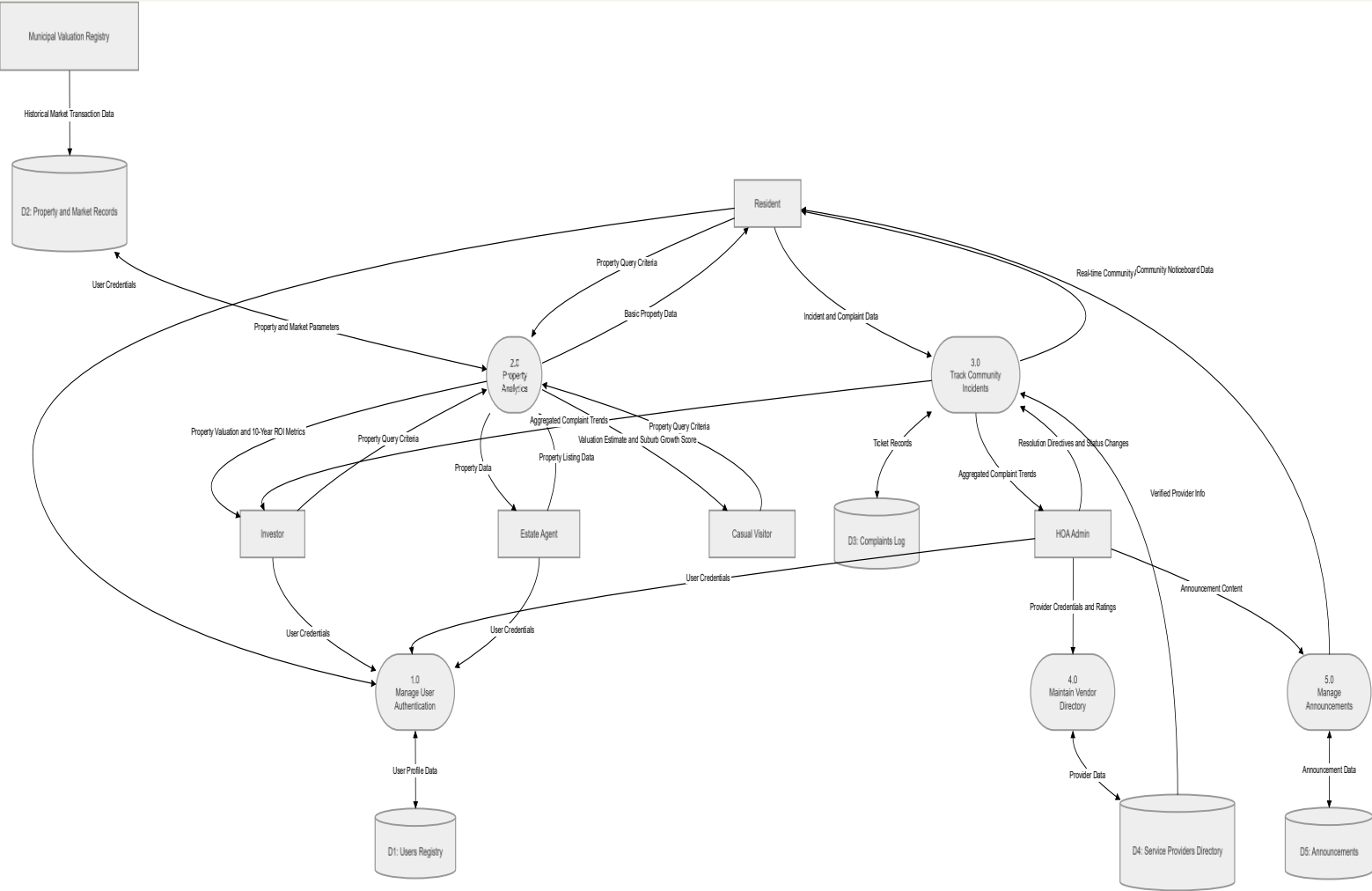
VALORA DATA MODELS

1. Context Level 0



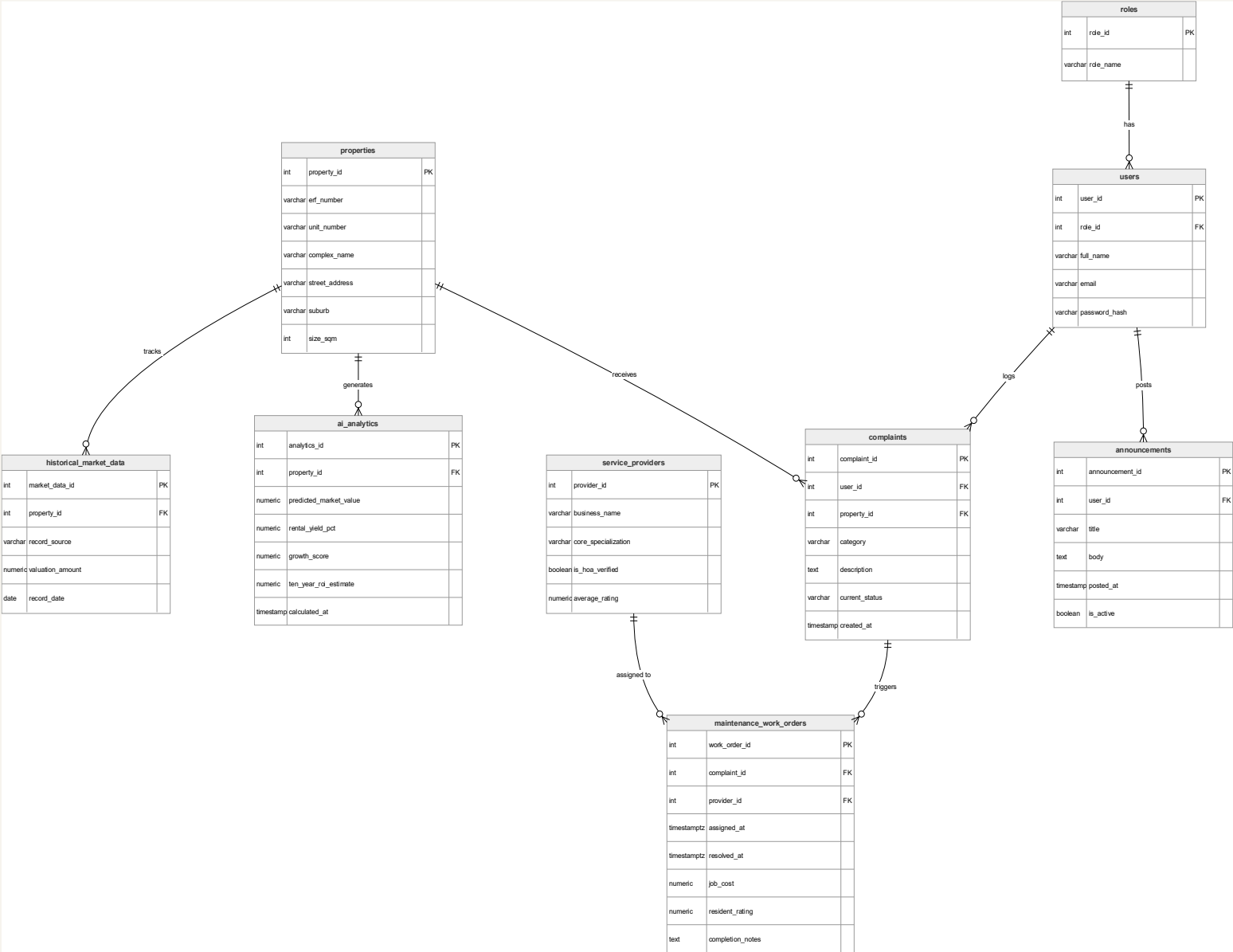
The context diagram provides a high-level view of the Valora system, showing external actors and the information exchanged between them and the system.

2. Data Flow Diagram



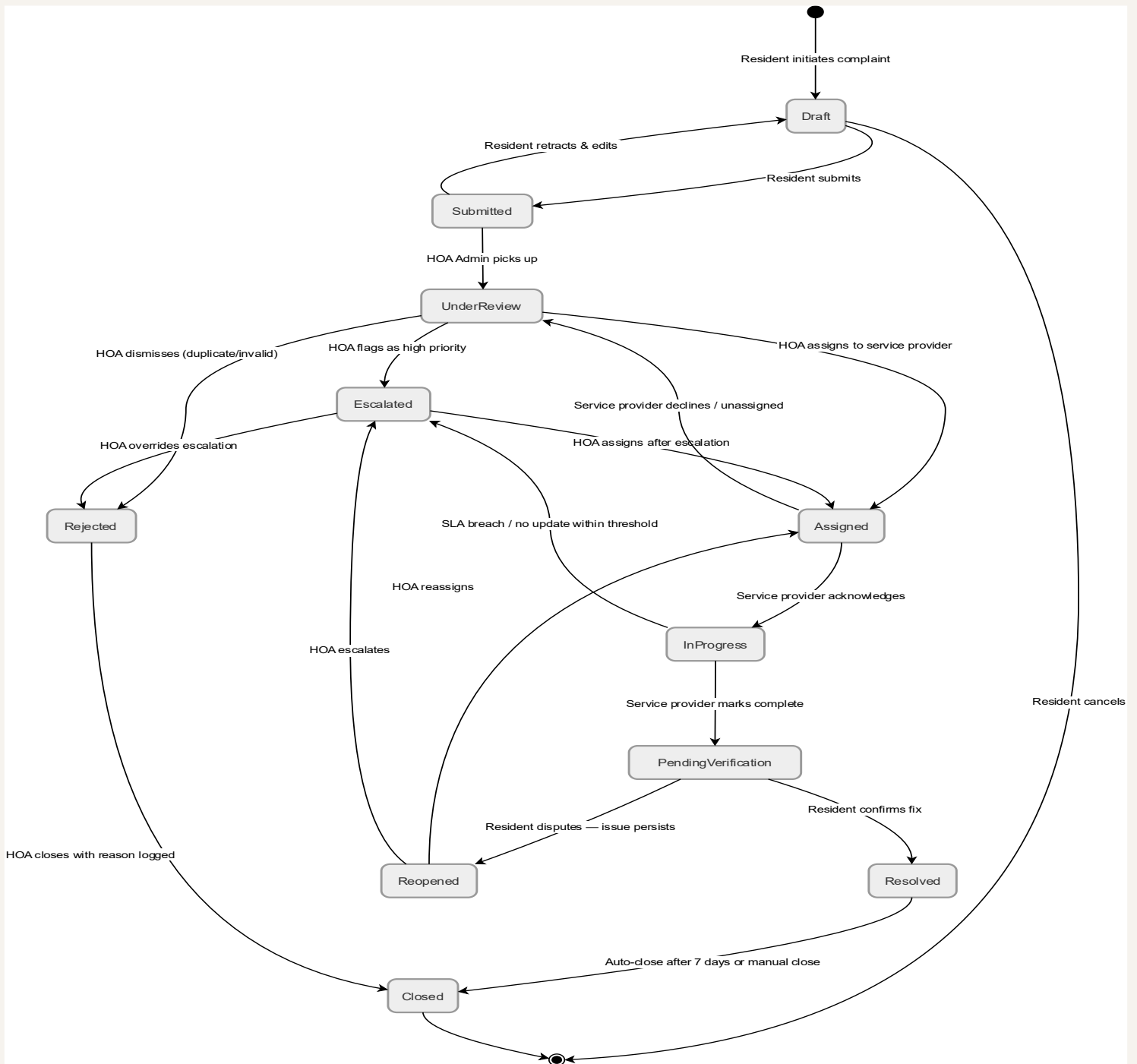
The DFD illustrates how data moves through the system, including processes, data stores, inputs, and outputs.

3. Entity Relationship Diagram



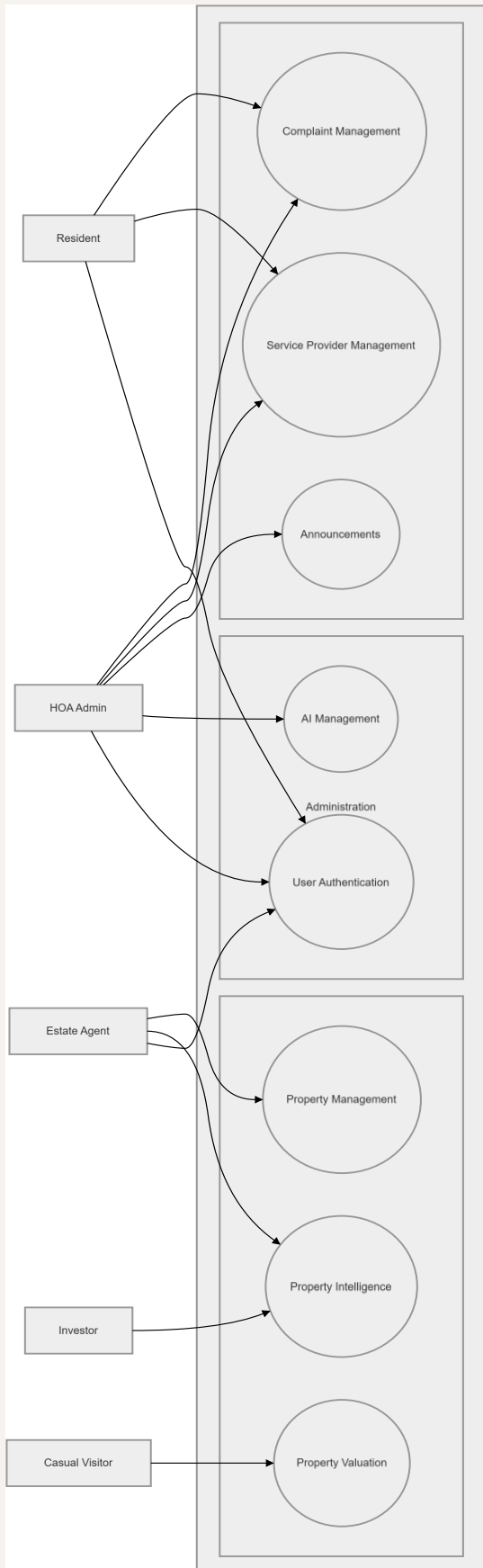
The ERD represents the system's database structure by showing entities, attributes, and relationships used to store and manage data.

4. State Transition



The state transition diagram traces the lifecycle of a complaint within the Valora system, mapping each possible state and the events or actions that trigger transitions between them.

5. Use-Case Diagram



The use case diagram identifies system users and the main functions available to each actor within the Valora platform.